

Neurodevelopmental outcome of fetal isolated ventricular asymmetry without dilation: a cohort study

R. MEYER^{1,2} , O. BAR-YOSEF^{2,3}, E. BARZILAY^{1,2} , D. HOFFMAN², S. TOUSSIA-COHEN², E. ZVI², R. ACHIRON^{1,2} and E. KATORZA^{1,2} 

¹Department of Obstetrics and Gynecology, The Chaim Sheba Medical Center, Ramat-Gan, Israel; ²Sackler School of Medicine, Tel Aviv University, Tel Aviv-Yafo, Israel; ³Pediatric Neurology Unit, The Chaim Sheba Medical Center, Ramat-Gan, Israel

KEYWORDS: fetal brain; MRI; neurodevelopmental outcome; ultrasound; ventricular asymmetry

ABSTRACT

Objective Fetal isolated ventricular asymmetry (IVA) is a relatively common finding in pregnancy, but data regarding its effect on neurodevelopmental outcome are scarce and founded principally on ultrasound-based studies. The purpose of this study was to assess the neurodevelopmental outcome of IVA cases in a magnetic resonance imaging (MRI)-based study.

Methods Cases referred for fetal brain MRI as part of the assessment of IVA without ventriculomegaly (lateral ventricular atrial diameter ≤ 10 mm), identified during routine ultrasound examination, were assessed for possible inclusion. Asymmetry was defined as a difference in width of ≥ 2 mm between the two lateral ventricles. Forty-three cases were included in the study group and compared with a control group of 94 normal cases without IVA. Children were assessed at ages 13–74 months using the Vineland-II Adaptive Behavior Scales (VABS-II).

Results VABS-II scores were within normal range. There was no significant difference in composite VABS-II score between the study and control groups (106.5 vs 108.0; $P = 0.454$). VABS-II scores did not differ between the groups when matched for gender and age at VABS-II interview (109.6 in study group vs 107.8 in control group; $P = 0.690$).

Conclusion In cases of IVA without ventriculomegaly on MRI, neurodevelopmental test scores were normal and did not differ from cases without IVA. Copyright © 2018 ISUOG. Published by John Wiley & Sons Ltd.

INTRODUCTION

Fetal lateral ventricular asymmetry is a common sonographic finding in prenatal examinations. The

definition of asymmetry is inconsistent but it has been defined as a difference in width of ≥ 2 mm between the two lateral ventricles in ultrasound- and magnetic resonance imaging (MRI)-based studies^{1–3}.

Data are limited regarding clinical outcome in cases of isolated ventricular asymmetry (IVA) without ventriculomegaly. Fetal ventriculomegaly has been defined previously as dilation of the lateral ventricular atrium to a width of > 10 mm^{4–6}. Moreover, no previous study has described the outcome of cases with ventricular asymmetry based on fetal MRI. Only three studies, two of them assessing the same group of subjects, have addressed specifically the neurodevelopmental outcome of IVA without dilation assessed using ultrasound, and found reassuring results, although these studies included relatively small groups of subjects^{1,3,7}.

It has been demonstrated that fetal MRI can detect more anomalies in cases of seemingly isolated ventriculomegaly or ventricular asymmetry, and allows improved resolution and direct visualization of both sides of the fetal brain^{8–13}. It has been used increasingly as an integral part of prenatal assessment for diverse indications¹⁴.

Cases of IVA without dilation are referred frequently to our tertiary center. The paucity of data regarding the postnatal clinical outcome of these cases currently prevents us and others from being able to provide thorough consultation.

In this study, we aimed to assess the neurodevelopmental outcome of fetuses with IVA by postnatal neurodevelopmental examinations in a MRI-based study in which MRI was used to further confirm the absence of additional central nervous system (CNS) findings. Our hypothesis was that the neurodevelopmental outcome of these cases would be normal. To the best of our knowledge, this is the first study to use MRI to explore this question.

Correspondence to: Dr R. Meyer, Department of Obstetrics and Gynecology, The Chaim Sheba Medical Center, Tel-Hashomer 52621, Israel (e-mail: raananmeir@gmail.com)

Accepted: 23 March 2018

METHODS

This was a cohort study of cases with isolated ventricular asymmetry without ventriculomegaly documented on MRI examination, performed at a single tertiary center. All singleton pregnancies that underwent fetal brain MRI at our center between July 2011 and January 2015 due to ventricular asymmetry without ventriculomegaly detected during routine ultrasound examination were assessed for possible inclusion in the study. As part of the investigation, these cases were offered a detailed anatomical ultrasound examination, fetal brain MRI, infection screening, fetal echocardiogram, genetic counseling and amniocentesis, as well as comparative genomic hybridization (CGH).

The inclusion criterion was IVA detected on ultrasound and confirmed on MRI examination. Exclusion criteria were: ventriculomegaly, fetal head circumference > 2 SD above the mean, CNS anomaly, non-CNS congenital anomaly, suspected intrauterine growth restriction (IUGR) or macrosomia, moderate–severe polyhydramnios and multiple pregnancy. Cases were also excluded that had fetal cytomegalovirus or toxoplasma infection, abnormal genetic tests or preterm delivery. Cases that did not meet the criteria for ventricular asymmetry without ventriculomegaly on MRI examination (due mainly to progression to ventriculomegaly) were excluded from the study, as were cases diagnosed with CNS anomalies on MRI examination. Cases were included that had minor additional imaging findings on fetal body ultrasound or fetal brain MRI examination that were considered not clinically significant.

Ventricular asymmetry was defined as a difference in width of ≥ 2 mm between the lateral ventricles on ultrasound and MRI examination. This value was selected based on previous ultrasound and MRI studies using this definition^{1–3}. Lateral ventricular width of 10 mm is the widely accepted cut-off for the definition of normal ventricular width, based mostly on ultrasound studies^{5,12,15–18}. Previous studies have also used this definition for MRI examinations and have found close agreement between the two modalities^{19,20}. We therefore included cases with ventricular width of ≤ 10 mm on ultrasound and MRI examinations.

Targeted ultrasound examinations (neurosonograms) were performed at our center by a fetal sonographer. Measurement of ventricular atrial width was performed in the transventricular plane. The calipers were positioned at the level of the glomus of the choroid plexus, touching the inner edge of the ventricle wall at its widest part and aligned perpendicular to the long axis of the ventricle, according to the International Society of Ultrasound in Obstetrics and Gynecology guidelines⁶.

Fetal brain MRI was performed using a 1.5-T system (Optima 1.5 T; GE Healthcare Ultrasound, Milwaukee, WI, USA), as described previously⁸. Single-shot fast spin-echo T2-weighted sequences in orthogonal planes were performed using the following parameters: section thickness 3–4 mm; no gap; flexible coil (eight-channel cardiac coil); matrix 320×224 ; echo time (TE) 90 ms;

and repetition time (TR) 1298 ms. The field of view was determined by the size of the fetal head: 24 cm for the smaller fetuses and 30 cm for the larger fetuses. T1 fast-spoiled gradient-echo sequences were performed only in the axial plane with a larger field of view (400 mm), with section thickness 4 mm, gap 0.5 mm, TR 160 ms and TE 2.3 ms. Measurements of the lateral ventricle widths were obtained in the coronal plane at the level of the ventricles, at the mid-height of the ventricle, with good visibility of the choroid plexuses¹⁹.

All newborns were assessed clinically by a neonatologist during the first 2 days after delivery.

Data were obtained from medical records and telephone interview with the children's parents, and included maternal medical and obstetric history, routine pregnancy follow-up examinations, TORCH status and perinatal history. Information on fetal echocardiogram, amniocentesis and CGH was collected at the time of these examinations. In addition, records of neurological clinical follow-up, including cases with additional genetic tests, were collected retrospectively.

Children were assessed using the Vineland-II Adaptive Behavior Scales (VABS-II), a structured parent interview assessing four different scales of performance: communication, daily living skills, socialization and motor skills²¹. The VABS-II is the most frequently used measure of adaptive behaviors such as communication, participation in community activities and development of relationships, and is considered a reliable and valid assessment tool^{22–24}. The score derived from this instrument has a mean of 100 ± 15 . A cut-off of 85 was used to denote normal *vs* delayed function. The published Israeli version of the VABS-II was administered, but no Israeli norms are available for the full age range. Because there was no reason to believe that children from Israel and the USA would develop mental and motor skills differently, we used USA norms²⁵. The study group was compared with a control group of neonates born at our institution between March 2011 and March 2015. The control group comprised cases with normal pregnancy follow-up, including normal second-trimester ultrasonographic anatomical scans, without IVA or other CNS abnormality. Other inclusion and exclusion criteria were similar to those of the study group.

To further compare neurodevelopmental outcomes that are dependent on gender and the children's age at VABS-II completion, cases and controls were matched according to these two variables. The criterion for matching was up to 4 months' difference in age at VABS-II.

Statistical analyses were conducted using SPSS Statistics v.23 (IBM Corp., Armonk, NY, USA). Histograms and Q–Q plots were used to assess for normality. Continuous variables were described as mean $\pm$ standard deviation or median and interquartile range. Continuous variables were compared using Student's *t*-test for independent samples or the Mann–Whitney *U*-test. The chi-square or Fisher's exact test was used for comparison of categorical variables. The matched groups were compared using a paired samples *t*-test or the Wilcoxon signed ranks test.

All statistical tests were two sided. Significance was set at $P < 0.05$.

The study was approved by the local institutional review board of the Sheba Medical Center. Informed consent was requested via letters sent to eligible cases prior to telephone calls and approved upon the VABS-II interview.

RESULTS

One hundred and seventy-five cases were identified that underwent fetal MRI examination due to ventricular asymmetry without ventriculomegaly on ultrasound examination. After excluding non-isolated cases of ventricular asymmetry on ultrasound examination and multiple pregnancies, we reviewed the MRI of 136 cases (Figure 1). Three cases were excluded due to suspected major CNS abnormality on MRI, 74 cases due to ventriculomegaly on MRI, and five because no asymmetry was diagnosed on MRI.

A total of 54 singleton pregnancies with IVA without dilation on MRI were eligible to participate in the VABS-II interview. Of these, five declined to participate,

two were lost to follow-up and, in four cases, we encountered technical difficulties precluding completion of the interview. The final study group for statistical analysis included 43 cases.

Ninety-eight cases were identified as adequate for inclusion in the control group. Of these, 94 agreed to participate and completed the VABS-II interview.

Maternal medical history, pregnancy and neonatal characteristics, as well as the children's age at VABS-II are depicted in Tables 1 and 2. Rate of previous delivery, fetal gender, gestational age at delivery, birth weight and children's age at VABS-II test differed significantly between the groups; 33/43 children in the study group were male compared with 52/94 in the control group (76.7% *vs* 55.3%; $P = 0.016$). The children's median age during the VABS-II test was 26.0 months (interquartile range (IQR), 21.0–28.0; range, 13–68 months) in the study group and 51.2 (IQR, 36.9–60.8; range, 26–74 months) in the control group ($P < 0.001$) (Table 2). TORCH results were not available in 4/43 (9.3%) cases in the study group. Amniocentesis was performed in 19/43 (44.2%) cases and CGH in 4/43 (9.3%). The results were considered normal in all cases.

In 6/43 cases in the study group, minor non-CNS findings were diagnosed on ultrasound, and in 6/43 cases minor CNS findings were diagnosed on MRI (Table 3). These cases were adequate for inclusion

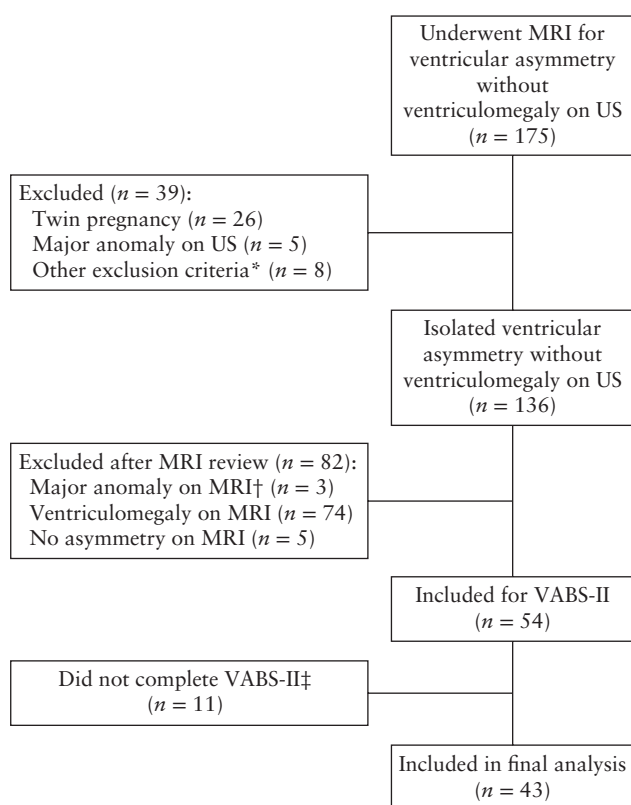


Figure 1 Flowchart summarizing selection of cases of isolated ventricular asymmetry assessed for neurodevelopmental outcome. *Congenital cytomegalovirus infection ($n = 2$), abnormal karyotype ($n = 1$), suspected intrauterine growth restriction ($n = 4$), suspected macrosomia ($n = 2$) (one case met two exclusion criteria). †Suspected brain infarct ($n = 1$), suspected lissencephaly ($n = 1$), suspected polymicrogyria ($n = 1$). ‡Declined to participate ($n = 5$), lost to follow-up ($n = 2$), technical problems in completing interview ($n = 4$). MRI, magnetic resonance imaging; US, ultrasound; VABS-II, Vineland-II Adaptive Behavior Scales.

Table 1 Patients, pregnancy and neonatal characteristics of cases of isolated ventricular asymmetry (IVA) and controls

Parameter	IVA (n = 43)	Controls (n = 94)	P
Maternal age (years)	30.8 ± 4.4	32.3 ± 5	0.090
BMI (kg/m ²)*	28.7 ± 3.6	28.0 ± 3.9	0.522
Any medical history	7/43 (16.3)	21/94 (22.3)	0.414
Family history of CNS abnormality†	6/43 (14.0)	ND	
Previous pregnancy	30/43 (69.8)	69/94 (73.4)	0.659
Previous delivery	21/43 (48.8)	64/94 (68.1)	0.031
Mode of conception			0.850
Spontaneous	41/43 (95.3)	87/94 (92.6)	
IUI	1/43 (2.3)	2/94 (2.1)	
In-vitro fertilization	1/43 (2.3)	5/94 (5.3)	
Fetal gender			0.016
Male	33/43 (76.7)	52/94 (55.3)	
Female	10/43 (23.3)	42/94 (44.7)	
Mode of delivery			0.152
Normal vaginal	25/34 (73.5)	76/94 (80.9)	
Cesarean	6/34 (17.6)	16/94 (17.0)	
Vacuum	1/34 (2.9)	2/94 (2.1)	
Forceps	2/34 (5.9)	0/94 (0)	
GA at delivery (weeks)	39.4 ± 1.5	38.8 ± 1.3	0.049
Birth weight (g)	3347.9 ± 409.7	3166.3 ± 372.8	0.011
1-min Apgar score ≤ 7	1/38 (2.6)	1/94 (1.1)	0.494
5-min Apgar score ≤ 7	0/38 (0)	0/94 (0)	

Data are given as mean ± SD or n/N (%). *Missing data for 28 cases in study group and 15 in control group. †Including congenital structural central nervous system (CNS) abnormalities, microcephaly, macrocephaly, cerebral palsy and learning disorders, but not including asymmetry. BMI, body mass index; GA, gestational age; IUI, intrauterine insemination; ND, no data.

Table 2 Vineland-II Adaptive Behavior Scales (VABS-II) scores of children with isolated ventricular asymmetry (IVA) and controls

	IVA (n = 43)	Controls (n = 94)	P
Age at VABS-II (months)	26.0 (21.0–28.0)	51.2 (36.9–60.8)	<0.001
VABS-II score			
Composite score	106.5 ± 10.9	108.0 ± 10.7	0.454
Communication	108.9 ± 12.2	108.8 ± 12.6	0.975
Daily living skills	109.2 ± 11.4	108.2 ± 10.0	0.631
Social skills	100.4 ± 13.4	106.3 ± 11.1	0.009
Motor skills	100.9 ± 10.9	104.3 ± 11.6	0.112

Data given as median (interquartile range) or mean ± SD.

Table 3 Imaging details of 43 cases of isolated ventricular asymmetry

Parameter	Value
GA at US prior to MRI	29.0 (26.9–30.8)‡
Ventricle width on US (mm)	
Smaller	6.0 (5.0–6.2)§
Larger	9.8 (9.0–10.0)§
GA at MRI (weeks)	32.0 (31.0–33.0)
Ventricle width on MRI (mm)	
Smaller	7.0 (6.0–7.0)
Larger	10.0 (9.0–10.0)
Minor non-CNS US findings*	6/43 (14.0)
Minor CNS MRI findings†	6/43 (14.0)

Data are given as median (interquartile range) or *n/N* (%). *One bilateral mild pyelectasis with mild polyhydramnios, one unilateral mild pyelectasis, one clenched hands, one liver calcifications, one small ventriculoseptal defect, one nuchal cyst. †Two mega cisterna magna, three occipitofrontal diameter ≥ 95th percentile, one prominent germinal matrix. ‡Data missing for three cases. §Data missing for four cases. CNS, central nervous system; GA, gestational age; MRI, magnetic resonance imaging; US, ultrasound.

Table 4 Vineland-II Adaptive Behavior Scales (VABS-II) scores for cases of isolated ventricular asymmetry (IVA) matched with controls by fetal gender and age at VABS-II

	IVA (n = 17)	Controls (n = 17)	P
Age at VABS-II (months)	36.35 ± 13.3	37.0 ± 13.1	0.266
Male gender	12/17 (70.6)	12/17 (70.6)	
Female gender	5/17 (29.4)	5/17 (29.4)	
VABS-II score			
Composite score	109.6 ± 11.6	107.8 ± 13.6	0.690
Communication	112.8 ± 13.0	107.9 ± 14.2	0.255
Daily living skills	111.9 ± 12.5	110.4 ± 13.2	0.745
Social skills	101.6 ± 17.3	103.2 ± 12.3	0.749
Motor skills	102.8 ± 11.9	105.6 ± 11.8	0.480

Data given as mean ± SD or *n/N* (%).

according to the criteria. In all other cases, imaging results were considered normal with no findings other than ventricular asymmetry. Median age at MRI examination was 32.0 (IQR, 31.0–33.0) months. The mean composite VABS-II score did not differ between the study and control groups (106.5 ± 10.9 *vs* 108.0 ± 10.7; *P* = 0.454;

Table 2). Within the different components of the VABS-II score, there was no significant difference between groups in communication, daily living skills and motor skills scores. However, the social skills score was significantly lower in the study group compared with the control group (100.4 *vs* 106.3; *P* = 0.009).

VABS-II scores were within normal range in both groups. There were no cases of abnormal composite VABS-II score (< 85) in either group. In 5/43 (11.6%) cases in the study group and 6/94 (6.4%) cases in the control group, one or two of the four VABS-II components were abnormal. Of note is that the proportion of additional CNS findings was relatively high among these cases: 3/5 in the study group and 2/6 in the control group.

When comparing cases of IVA with controls matched for gender and the child's age at VABS-II, there was no significant difference in composite VABS-II score (109.6 in the study group *vs* 107.8 in the control group; *P* = 0.690), or any of its components (Table 4). None of the cases needed further genetic evaluation after birth.

DISCUSSION

In this study, we investigated the neurodevelopmental outcome of fetuses diagnosed with IVA without dilation on MRI examination. The finding of 'asymmetric lateral brain ventricles' during routine sonographic evaluation is a frequent reason for referral of patients to our tertiary center.

Functional and structural asymmetry of the brain is considered a normal developmental phenomenon²⁶. It has been shown in ultrasound and MRI studies to be side dependent, with the left side being significantly larger than the right one in normal fetuses, newborns and adults^{27–29}. Kivilevitch *et al.* found a mean difference of 6.5% (0.35 mm) between the lateral ventricles on fetal ultrasound examinations²⁶. In a MRI-based study, Kyriakopoulou *et al.* found a mean difference of 17.9% in ventricular volumes in normal fetuses²⁷. However, the clinical significance of large asymmetry (≥ 2 mm diameter difference) is unclear^{1,3}. Furthermore, this question has not been investigated in a MRI-based study.

We found no difference in the neurodevelopmental score of cases with IVA compared with infants of pregnancies with normal follow-up examinations. Moreover, within the 43 cases of IVA, we found no abnormal VABS-II scores. The only significant difference between the study and control groups was in the social skills component of the VABS-II, which did not affect the composite score (Table 2).

We performed a matching analysis to control for the significant differences between groups in age at VABS-II and gender, in which the significant difference in social skills found in the primary analysis was no longer present (Table 4). It is possible that social skills differed in the primary analysis due to a higher proportion of males in the study group. It should be noted, however, that the VABS-II is structured to yield age-adjusted scores²¹. Accordingly, age is not expected to affect VABS-II scores.

To the best of our knowledge, only three studies have investigated specifically outcomes in cases with IVA without ventriculomegaly. Two of them examined the same group of subjects at different ages. Fetal CNS evaluation was performed only via ultrasound in these studies.

The study by Achiron *et al.* defined lateral ventricular asymmetry as a difference in width of ≥ 2.4 mm and found normal developmental outcome in all 17 cases at 6 months of age¹. The study by Sadan *et al.* defined asymmetry as a difference of > 2 mm in width between the lateral ventricles, and found similar mental and motor scores and a significant risk for behavioral abnormalities in the ventricular asymmetry group, comprising 21 children, compared with the control group of 20 children³. A follow-up study on subjects from the same group at the age of 9–11 years showed no difference in mental development compared with the general population⁷. Our results are in accordance with those of these studies and further suggest that IVA, diagnosed on MRI, does not affect neurodevelopmental outcome.

In addition to the current study being the largest to assess the neurodevelopmental outcome of fetuses diagnosed with IVA, it is also the first to use fetal MRI as part of IVA assessment. MRI is considered an accurate tool and has been shown to detect additional findings in cases of seemingly isolated ventriculomegaly or asymmetry suspected on ultrasound examination^{8,12}. As depicted in Figure 1, suspected major additional findings were diagnosed on MRI in three of 57 cases with IVA suspected on ultrasound examination. Yet, no conclusion can be drawn on the necessity of MRI in cases of IVA suspected on ultrasound examination, as the current study was not structured to address this question, and it should be researched in the future in specifically designed studies. In addition, of the 136 cases with ventricular asymmetry without ventriculomegaly on ultrasound examination, 74 were eventually diagnosed with ventriculomegaly on MRI, and five did not have ventricular asymmetry. These differences are most likely due to progression of ventricular dilation, as the median gestational age at ultrasound examination was 29.0 weeks compared with 32.0 at MRI. Other possible explanations are interobserver differences or, perhaps, disagreement between the two modalities.

This study has several limitations. First, we used MRI to confirm the diagnosis of IVA without ventriculomegaly. The definition of asymmetry is based mainly on ultrasound studies and current literature regarding MRI lateral ventricles reference values is lacking^{1–3}. However, ultrasound reference values have been used in MRI studies and previous studies have reported good agreement between the two modalities^{19,20}. A second limitation is the use of the VABS-II based on USA norms in an Israeli population. However, there is no reason to believe that Israeli and US children would develop mental and motor skills differently. Furthermore, the comparison with an equivalent Israeli control group serves as validation of our results. Third, although all patients in the study group

were offered thorough prenatal diagnosis, TORCH, amniocentesis and CGH results were not available in all cases. In addition, data regarding the above stated examinations in the control group are missing. Thus, it is possible that some cases had an underlying infectious or genetic abnormality. However, the impact of undiagnosed cases of genetic or congenital infections on VABS-II scores of both groups is probably low, as both groups had normal second-trimester anatomical scans, no cases of trisomy were detected after birth in either group, the prevalence of congenital infections in our population is relatively low and VABS-II scores were normal in both groups^{30–33}. Fourth, the subanalysis matching for gender and age at VABS-II was not powered to determine a statistically significant difference between the two groups. Therefore, we cannot rule out the possibility that our results were biased due to differences between the study and control populations.

This study presents several strengths. First, it is the largest study investigating outcome of IVA. Second, the IVA group was compared with a large control group with detailed pregnancy information from our computerized database. Third, the use of MRI and the thorough pregnancy follow-up enabled us to detect a highly selected study group of isolated ventricular asymmetry without ventriculomegaly.

In conclusion, in cases of IVA undergoing MRI, neurodevelopmental scores were normal and did not differ from cases without ventricular asymmetry.

REFERENCES

- Achiron R, Yagel S, Rotstein Z, Inbar O, Mashiach S, Lipitz S. Cerebral lateral ventricular asymmetry: is this a normal ultrasonographic finding in the fetal brain? *Obstet Gynecol* 1997; 89: 233–237.
- Barzilay E, Bar-Yosef O, Dorembus S, Achiron R, Katorza E. Fetal brain anomalies associated with ventriculomegaly or asymmetry: An MRI-based study. *AJNR Am J Neuroradiol* 2017; 38: 371–375.
- Sadan S, Malinger G, Schweiger A, Lev D, Lerman-Sagie T. Neuropsychological outcome of children with asymmetric ventricles or unilateral mild ventriculomegaly identified in utero. *BJOG* 2007; 114: 596–602.
- Almog B, Gamzu R, Achiron R, Fainaru O, Zalel Y. Fetal lateral ventricular width: what should be its upper limit? A prospective cohort study and reanalysis of the current and previous data. *J Ultrasound Med* 2003; 22: 39–43.
- Cardoza JD, Goldstein RB, Filly RA. Exclusion of fetal ventriculomegaly with a single measurement: the width of the lateral ventricular atrium. *Radiology* 1988; 169: 711–714.
- International Society of Ultrasound in Obstetrics & Gynecology Education Committee. Sonographic examination of the fetal central nervous system: guidelines for performing the 'basic examination' and the 'fetal neurosonogram'. *Ultrasound Obstet Gynecol* 2007; 29: 109–116.
- Atad-Rapoport M, Schweiger A, Lev D, Sadan-Strul S, Malinger G, Lerman-Sagie T. Neuropsychological follow-up at school age of children with asymmetric ventricles or unilateral ventriculomegaly identified in utero. *BJOG* 2015; 122: 932–938.
- Gat I, Hoffmann C, Shashar D, Yosef OB, Konen E, Achiron R, Brandt B, Katorza E. Fetal brain MRI: Novel classification and contribution to sonography. *Ultraschall Med* 2016; 37: 176–184.
- Morris JE, Rickard S, Paley MN, Griffiths PD, Rigby A, Whitby EH. The value of in-utero magnetic resonance imaging in ultrasound diagnosed foetal isolated cerebral ventriculomegaly. *Clin Radiol* 2007; 62: 140–144.
- Benacerraf BR, Shipp TD, Bromley B, Levine D. What does magnetic resonance imaging add to the prenatal sonographic diagnosis of ventriculomegaly? *J Ultrasound Med* 2007; 26: 1513–1522.
- Kandula T, Fahey M, Chalmers R, Edwards A, Shekleton P, Teoh M, Clark J, Goergen SK. Isolated ventriculomegaly on prenatal ultrasound: what does fetal MRI add? *J Med Imaging Radiat Oncol* 2015; 59: 154–162.
- Salomon LJ, Ouahba J, Delezoide AL, Vuillard E, Oury JF, Sebag G, Garel C. Third-trimester fetal MRI in isolated 10- to 12-mm ventriculomegaly: is it worth it? *BJOG* 2006; 113: 942–947.
- Vazquez E, Mayolas N, Delgado I, Higuera T. Fetal neuroimaging: US and MRI. *Pediatr Radiol* 2009; 39 (Suppl 3): 422–435.

14. Yin S, Na Q, Chen J, Li-Ling J, Liu C. Contribution of MRI to detect further anomalies in fetal ventriculomegaly. *Fetal Diagn Ther* 2010; 27: 20–24.
15. Farrell TA, Hertzberg BS, Kliewer MA, Harris L, Paine SS. Fetal lateral ventricles: reassessment of normal values for atrial diameter at US. *Radiology* 1994; 193: 409–411.
16. Hilpert PL, Hall BE, Kurtz AB. The atria of the fetal lateral ventricles: a sonographic study of normal atrial size and choroid plexus volume. *AJR Am J Roentgenol* 1995; 164: 731–734.
17. Heiserman J, Filly RA, Goldstein RB. Effect of measurement errors on sonographic evaluation of ventriculomegaly. *J Ultrasound Med* 1991; 10: 121–124.
18. Alagappan R, Browning PD, Laorr A, McGahan JP. Distal lateral ventricular atrium: reevaluation of normal range. *Radiology* 1994; 193: 405–408.
19. Perlman S, Shashar D, Hoffmann C, Yosef OB, Achiron R, Katorza E. Prenatal diagnosis of fetal ventriculomegaly: Agreement between fetal brain ultrasonography and MR imaging. *AJNR Am J Neuroradiol* 2014; 35: 1214–1218.
20. Garel C. *MRI of the Fetal Brain: Normal Development and Cerebral Pathologies*. Springer-Verlag: Berlin, 2004; 201–216.
21. Sparrow S, Cicchetti D, Balla D. *Vineland Adaptive Behavior Scales* (2nd edn). Circle Pines, MN: American Guidance Service Publishing, 2005.
22. Pugliese CE, Anthony L, Strang JF, Dudley K, Wallace GL, Kenworthy L. Increasing adaptive behavior skill deficits from childhood to adolescence in autism spectrum disorder: role of executive function. *J Autism Dev Disord* 2015; 45: 1579–1587.
23. Moler FW, Silverstein FS, Dean JM. Hypothermia after in-hospital cardiac arrest in children. *N Engl J Med* 2017; 376: 1696–1697.
24. Taverna L, Tremolada M, Bonichini S, Tosetto B, Basso G, Messina C, Pillon M. Motor skill delays in pre-school children with leukemia one year after treatment: Hematopoietic stem cell transplantation therapy as an important risk factor. *PLoS One* 2017; 12: e0186787.
25. Bar-Yosef O, Barzilay E, Dorembus S, Achiron R, Katorza E. Neurodevelopmental outcome of isolated ventriculomegaly: a prospective cohort study. *Prenat Diagn* 2017; 37: 764–768.
26. Kivilevitch Z, Achiron R, Zalel Y. Fetal brain asymmetry: in utero sonographic study of normal fetuses. *Am J Obstet Gynecol* 2010; 202: 359.e1–359.e8.
27. Kyriakopoulou V, Vatansever D, Davidson A, Patkee P, Elkommos S, Chew A, Martinez-Biarge M, Hagberg B, Damodaram M, Allsop J, Fox M, Hajnal JV, Rutherford MA. Normative biometry of the fetal brain using magnetic resonance imaging. *Brain Struct Funct* 2017; 222: 2295–2307.
28. Gilmore JH, Lin W, Prastawa MW, Looney CB, Vetsa YS, Knickmeyer RC, Evans DD, Smith JK, Hamer RM, Lieberman JA, Gerig G. Regional gray matter growth, sexual dimorphism, and cerebral asymmetry in the neonatal brain. *J Neurosci* 2007; 27: 1255–1260.
29. Shenton ME, Kikinis R, McCarley RW, Metcalf D, Tieman J, Jolesz FA. Application of automated MRI volumetric measurement techniques to the ventricular system in schizophrenics and normal controls. *Schizophr Res* 1991; 5: 103–113.
30. Franklin DM, Dror Z, Nishri Z. The prevalence and incidence of Toxoplasma antibodies in pregnant women. *Isr J Med Sci* 1993; 29: 285–286.
31. Schlesinger Y, Halle D, Eidelman AI, Reich D, Dayan D, Rudensky B, Raveh D, Branski D, Kaplan M, Shefer V, Miron D. Urine polymerase chain reaction as a screening tool for the detection of congenital cytomegalovirus infection. *Arch Dis Child Fetal Neonatal Ed* 2003; 88: F371–F374.
32. Schlesinger Y, Reich D, Eidelman AI, Schimmel MS, Hassanin J, Miron D. Congenital cytomegalovirus infection in Israel: screening in different subpopulations. *Isr Med Assoc J* 2005; 7: 237–240.
33. Levine H, Ankol OE, Rozhavski V, Davidovitch N, Aboudy Y, Zarka S, Balicer RD. Rubella seroprevalence in the first birth cohort reaching fertility age after 20 years of two dose universal vaccination policy in Israel. *Vaccine* 2012; 30: 7260–7264.